Supervised Learning for Medical Diagnosis:
A Comparative Analysis of Logistic Regression and Support Vector Machines on a Breast Cancer Dataset

Abstract

Supervised learning has become a fundamental approach in modern data-driven decision-making, particularly in high-stakes domains such as healthcare. This tutorial presents a detailed comparative analysis of logistic regression and support vector machines (SVM) applied to the Wisconsin Breast Cancer dataset. The study explores the complete supervised learning pipeline, including preprocessing, feature scaling, model training, and evaluation using metrics such as accuracy, confusion matrices, and ROC curves. Particular emphasis is placed on understanding how model complexity, kernel selection, and data characteristics influence performance. The results demonstrate that while simple linear models provide strong baselines, more flexible approaches such as RBF SVM can capture subtle patterns in the data and improve predictive performance. The tutorial also highlights the importance of interpretability, ethical considerations, and accessibility in medical machine learning applications.

1. Introduction

Machine learning has transformed the way complex data is analysed across a wide range of disciplines, from finance to healthcare. Among the different paradigms within machine learning, supervised learning remains one of the most widely used due to its ability to learn directly from labelled data. In supervised learning, a model is trained using input-output pairs, allowing it to learn a mapping function that can be used to predict outcomes for unseen data (Bishop, 2006).

In classification tasks, the goal is to assign each input to a predefined category. This is particularly relevant in medical diagnostics, where patient data can be used to determine whether a condition is present or absent. For example, based on measurements extracted from medical images, a model can be trained to classify tumours as either malignant or benign. Such applications are not only technically challenging but also carry significant real-world implications, as incorrect predictions can affect patient outcomes.

This tutorial focuses on applying supervised learning techniques to the problem of breast cancer classification using the Wisconsin Breast Cancer dataset. Two widely used models are explored in depth:

- **Logistic Regression**, a linear probabilistic classifier that provides interpretable outputs
- **Support Vector Machines (SVM)**, a powerful classification method capable of modelling both linear and non-linear relationships

In addition, different SVM kernels (linear, polynomial, and radial basis function) are examined to understand how model flexibility influences performance. The primary objective of this tutorial is not only to compare these models quantitatively but also to provide a clear conceptual understanding of their behaviour and limitations. This aligns with key learning outcomes in machine learning, including the ability to implement algorithms, evaluate performance, and critically analyse results.

2. Dataset Overview

The dataset used in this study is the **Wisconsin Breast Cancer (Diagnostic) dataset**, a widely recognised benchmark in machine learning research (Street et al., 1993). It contains 569 observations, each representing a tumour sample obtained from digitised images of fine needle aspirates of breast tissue.

Each sample is described by 30 numerical features that capture properties of the cell nuclei, including:

- Radius (mean distance from centre to boundary)
- Texture (variation in pixel intensity)
- Smoothness
- Concavity
- Symmetry

These features are designed to quantify subtle differences between malignant and benign tumours.

The target variable, labelled diagnosis, indicates whether the tumour is:

- **Malignant (M)** — cancerous and potentially life-threatening
- **Benign (B)** — non-cancerous

This labelled structure makes the dataset ideal for supervised learning, as the model can learn patterns associated with each class.

An important characteristic of this dataset is that it is relatively clean and well-structured, which allows the focus to remain on model behaviour rather than extensive data cleaning. However, preprocessing steps such as encoding and scaling are still necessary to ensure optimal performance.

3. Supervised Learning Pipeline

A key aspect of any machine learning task is the design of a robust and reproducible pipeline. This ensures that the results are reliable and can be generalised to new data.

3.1 Label Encoding

Since machine learning models operate on numerical data, the categorical labels must be converted into numeric form. In this case:

- Malignant (M) is encoded as 1
- Benign (B) is encoded as 0

This transforms the classification problem into a binary task, where the model predicts the probability of malignancy.

3.2 Train/Test Split

To evaluate model performance, the dataset is divided into two subsets:

- **Training set (80%)** — used to train the model
- **Testing set (20%)** — used to evaluate performance

Stratification is applied to ensure that both subsets maintain the same proportion of malignant and benign cases. This is particularly important in medical datasets, where class imbalance can bias the model.

3.3 Feature Scaling

The features in the dataset vary significantly in magnitude. For example, radius values may be in tens, while other features may be much smaller. Without scaling, models such as SVM may become biased towards features with larger numerical values.

To address this, standardization is applied so that each feature has:

- Mean = 0
- Standard deviation = 1

Feature scaling is especially critical for SVM, as the algorithm relies on distance calculations. It also improves the convergence of optimisation algorithms used in Logistic Regression.

4. Supervised Models

4.1 Logistic Regression

Logistic Regression is one of the simplest yet most effective classification algorithms. Despite its name, it is used for classification rather than regression. The model estimates the probability that a given input belongs to a particular class using a logistic (sigmoid) function.

The decision boundary produced by Logistic Regression is linear, meaning it attempts to separate the classes using a straight line (or hyperplane in higher dimensions). This simplicity makes it highly interpretable, as the coefficients associated with each feature indicate their influence on the prediction.

In medical applications, this interpretability is particularly valuable. Clinicians can understand which features contribute most strongly to a diagnosis, making the model more transparent and trustworthy.

However, the main limitation of Logistic Regression is that it cannot capture complex non-linear relationships unless feature engineering is applied.

4.2 Support Vector Machines (SVM)

Support Vector Machines are more sophisticated models that aim to find the optimal boundary that separates classes with the largest possible margin (Cortes & Vapnik, 1995). The key idea is to maximise the distance between the decision boundary and the closest data points, known as support vectors.

One of the most powerful aspects of SVM is the use of **kernel functions**, which allow the model to transform the data into a higher-dimensional space where it becomes easier to separate.

In this tutorial, three kernels are explored:

- **Linear Kernel** — similar to Logistic Regression but focuses on margin maximisation
- **Polynomial Kernel** — introduces curved decision boundaries
- **RBF (Radial Basis Function) Kernel** — highly flexible and capable of modelling complex patterns

The choice of kernel significantly affects model performance, making it an important aspect of SVM design.

5. Evaluation Metrics

Evaluating machine learning models requires more than just accuracy, particularly in medical applications.

The following metrics are used:

- **Accuracy** — proportion of correct predictions
- **Precision** — proportion of predicted positives that are correct
- **Recall** — proportion of actual positives correctly identified
- **F1 Score** — balance between precision and recall
- **Confusion Matrix** — detailed breakdown of predictions
- **ROC Curve and AUC** — performance across different thresholds

Why These Metrics Matter

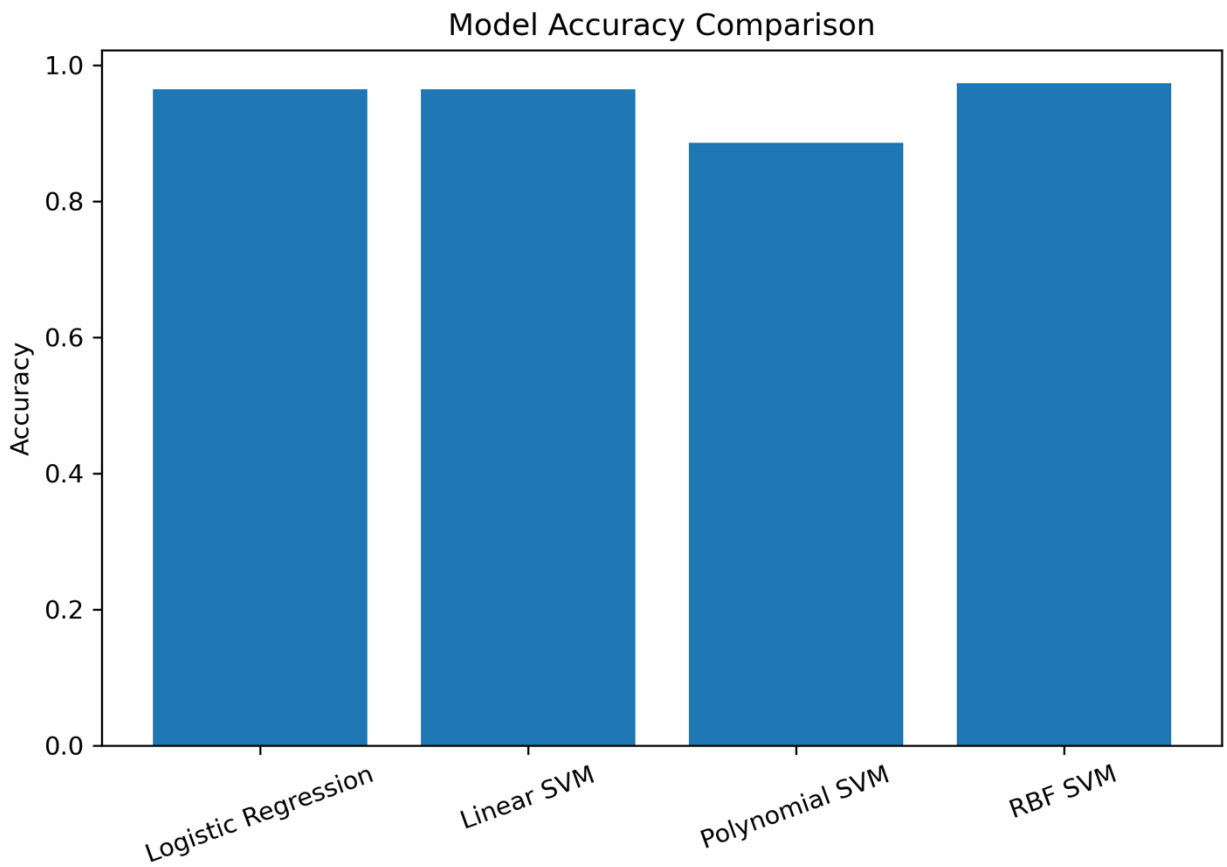
In medical diagnosis:

- A **false negative** (missing a malignant tumour) can delay treatment
- A **false positive** may cause unnecessary stress and procedures

Therefore, recall is often more critical than accuracy.

6. Results and Visualisation

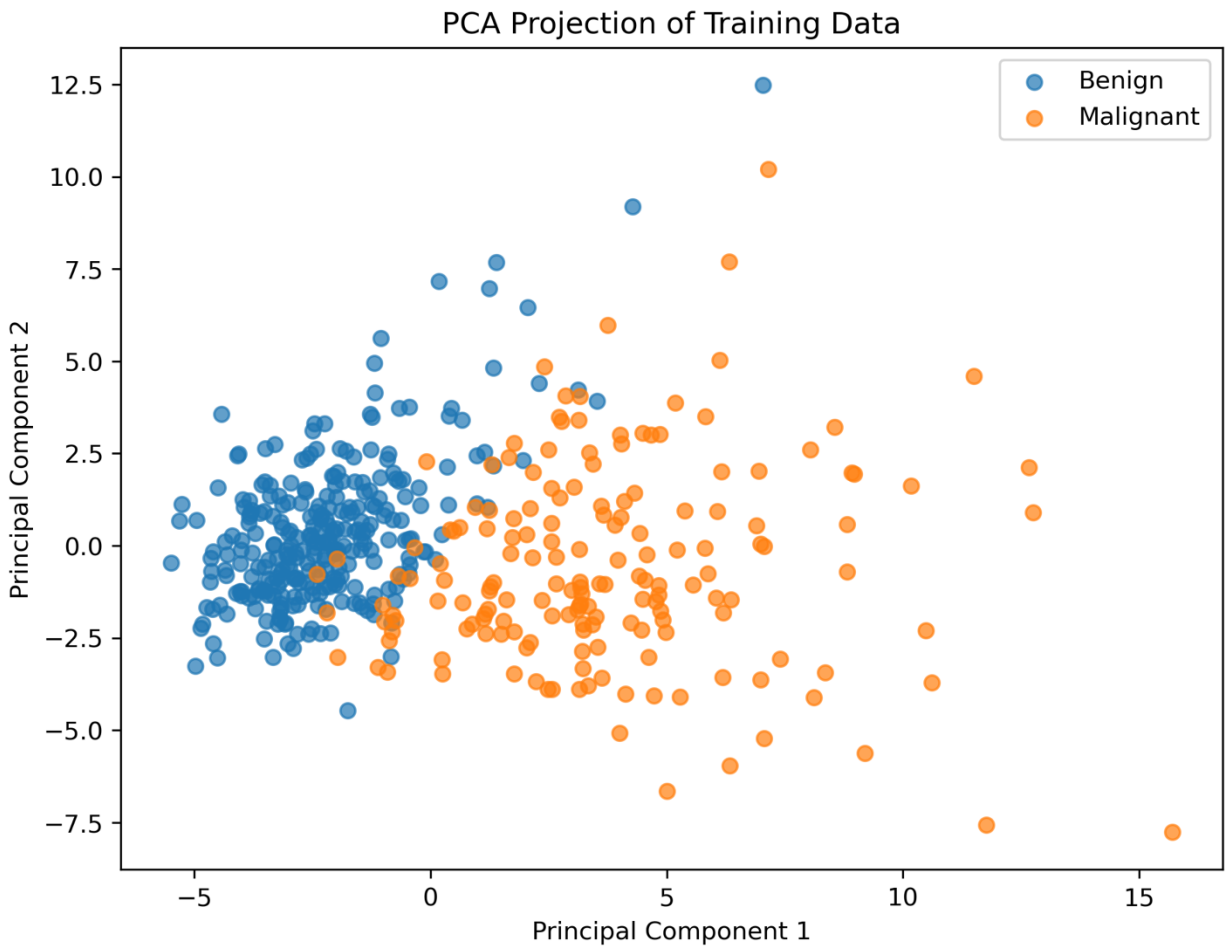
6.1 Accuracy Comparison



The results show that all models achieve high accuracy, indicating that the dataset is relatively well-structured. However, the RBF SVM consistently achieves the highest performance, followed closely by Logistic Regression and Linear SVM.

This suggests that while the data is largely linearly separable, there are subtle non-linear patterns that more flexible models can capture.

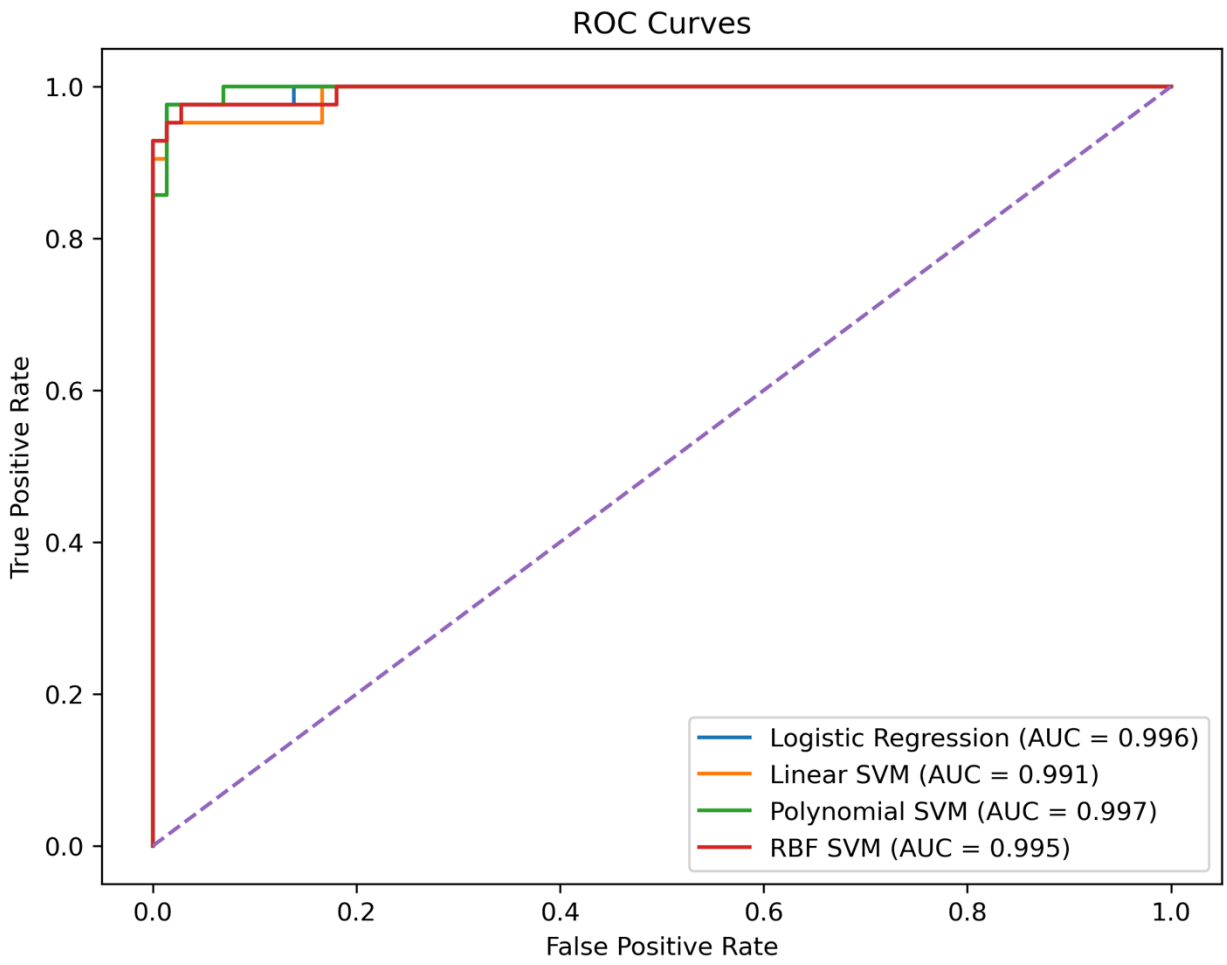
6.2 PCA Projection



Principal Component Analysis (PCA) is used to reduce the dimensionality of the dataset to two components for visualisation. The resulting scatter plot shows partial separation between malignant and benign samples.

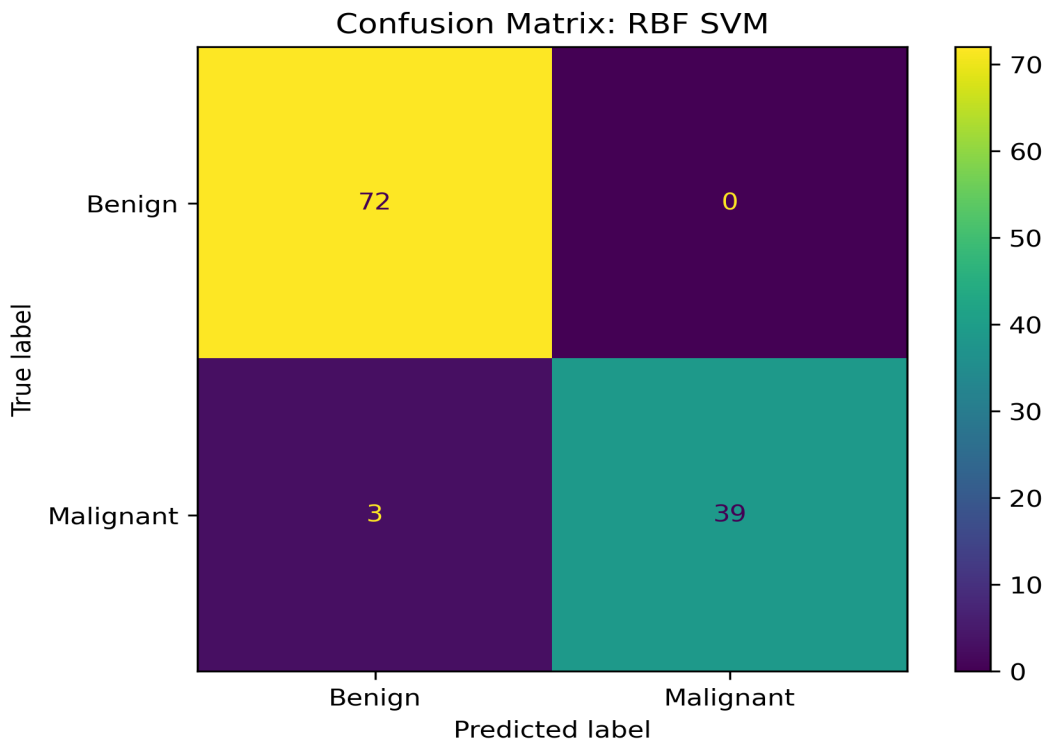
Although the classes overlap, there is still a visible structure that allows supervised models to distinguish between them effectively.

6.3 ROC Curves



The ROC curves provide a comprehensive view of model performance across different classification thresholds. All models achieve high AUC values, but the RBF SVM curve remains closest to the top-left corner, indicating superior performance.

6.4 Confusion Matrix



The confusion matrix reveals that most samples are correctly classified, with only a small number of misclassifications. This confirms the reliability of the models, particularly the RBF SVM.

7. Interpretation of Results

The comparison of models highlights several important insights:

- Logistic Regression performs well due to the near-linear structure of the data
- Linear SVM improves classification by maximising the margin
- Polynomial SVM introduces excessive complexity, leading to overfitting
- RBF SVM achieves the best balance between flexibility and generalisation

This demonstrates a key principle in machine learning:

Model complexity must match the complexity of the data.

From a clinical perspective, models with high recall for malignant cases are preferred, as missing a cancer diagnosis has serious consequences.

8. Accessibility Considerations

Accessibility is an important aspect of responsible data communication. All visualisations in this tutorial are designed with:

- Clear labels and titles
- Readable font sizes
- Colour-blind friendly palettes

Additionally, descriptive captions are provided to ensure compatibility with screen readers.

9. Ethical Considerations

The use of machine learning in healthcare raises important ethical concerns:

- Bias in training data can lead to unfair predictions
- Lack of transparency in complex models may reduce trust
- Incorrect predictions can have life-threatening consequences

Therefore, it is essential to ensure that models are carefully validated and used as decision-support tools rather than replacements for clinical judgement.

10. Conclusion

This tutorial demonstrated how supervised learning models can be applied to medical diagnosis using the breast cancer dataset.

The results show that while simple models such as logistic regression provide strong and interpretable baselines, more advanced models like RBF SVM can achieve higher accuracy by capturing complex patterns in the data.

Ultimately, the choice of model depends on the trade-off between accuracy, interpretability, and computational complexity. In high-stakes domains such as healthcare, this decision must be made carefully, taking into account both technical performance and ethical considerations.

11. References

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